

P.lagopus dbRDA

Andrés Delgado Galán

2026-02-24

In these analyses we evaluate how much epigenetic variability is explained by the herbivory treatments and their interaction. We analysed overall epigenetic variability and separately for each context. We used the methylation percent of each SMP (4531 in CpG, 3738 in CHH, 3738 in CHH) to construct an euclidean epigenetic distance matrix between samples (40) using `cluster::daisy`.

We conducted dbRDA using `vegan::capscale` with the epigenetic distance matrix as response and the herbivory treatments as predictors. To account for genetic variability, we summarized the SNP data (5501 SNPs) into principal components using PCA, and included the first PC (that explains 23.57% of the genetic variability) as a condition in a constrained dbRDA. We tested the significance of the model using a permutation test with 9999 permutations and obtained adjusted R2 values using the function `RsquareAdj` from the `vegan` package. We visualized the structure of the data in a 2-D space using the constrained ordination axes (RDA axes). We ran the following models for all contexts together and separately for each sequence context:

- 1. Epigenetic distance ~ Maternal herbivory
- 2. Epigenetic distance ~ Maternal herbivory * Real-time herbivory
- 3. Epigenetic distance ~ Maternal herbivory * Real-time herbivory + Condition (PC)
- 4. Epigenetic distance ~ PC

Table 1. Results of the distance-based redundancy analysis (dbRDA) showing the percentage of epigenetic variance (ED) explained by maternal environment (M), real-time herbivory simulation (S), and/or their interaction (M:S) across different genomic contexts.

Model	df	SumOfSqs	F	p_value	FDR	adj_R2
CpG						
ED ~ M	1	25490.10	2.043	<0.001	<0.001	0.026
ED ~ M * S	3	54053.47	1.456	<0.001	<0.001	0.034
ED ~ M * S + Condition(PC)	3	52476.74	1.439	<0.001	<0.001	0.033
ED ~ PC	1	21817.37	1.735	0.001	0.001	0.019
CHG						
ED ~ M	1	19906.94	1.372	0.019	0.019	0.009
ED ~ M * S	3	55413.15	1.289	0.004	0.008	0.022

The models include with and without adjusting for the variance explained by the genetic data (PC). PC, first principal component from genetic data (explaining 23.57% of the variation in scaled allele frequencies); df, degrees of freedom; F, value of the test statistic; FDR, false discovery rate; adj. R2, percent of variance explained, adjusted for multiple comparisons.

Model	df	SumOfSqs	F	p_value	FDR	adj_R2
ED ~ M * S + Condition(PC)	3	45213.99	1.136	0.009	0.011	0.010
ED ~ PC	1	61529.07	4.588	<0.001	0.002	0.084
CHH						
ED ~ M	1	23927.77	1.086	0.315	0.419	0.002
ED ~ M * S	3	103423.02	1.639	0.021	0.041	0.047
ED ~ M * S + Condition(PC)	3	47573.95	1.013	0.461	0.461	0.001
ED ~ PC	1	265173.23	16.916	<0.001	<0.001	0.290
All Contexts						
ED ~ M	1	69132.22	1.417	0.096	0.096	0.011
ED ~ M * S	3	211966.99	1.486	0.010	0.013	0.036
ED ~ M * S + Condition(PC)	3	145252.48	1.180	0.006	0.012	0.012
ED ~ PC	1	341936.51	8.216	<0.001	0.002	0.156

The models include with and without adjusting for the variance explained by the genetic data (PC). PC, first principal component from genetic data (explaining 23.57% of the variation in scaled allele frequencies); df, degrees of freedom; F, value of the test statistic; FDR, false discovery rate; adj. R2, percent of variance explained, adjusted for multiple comparisons.

CpG

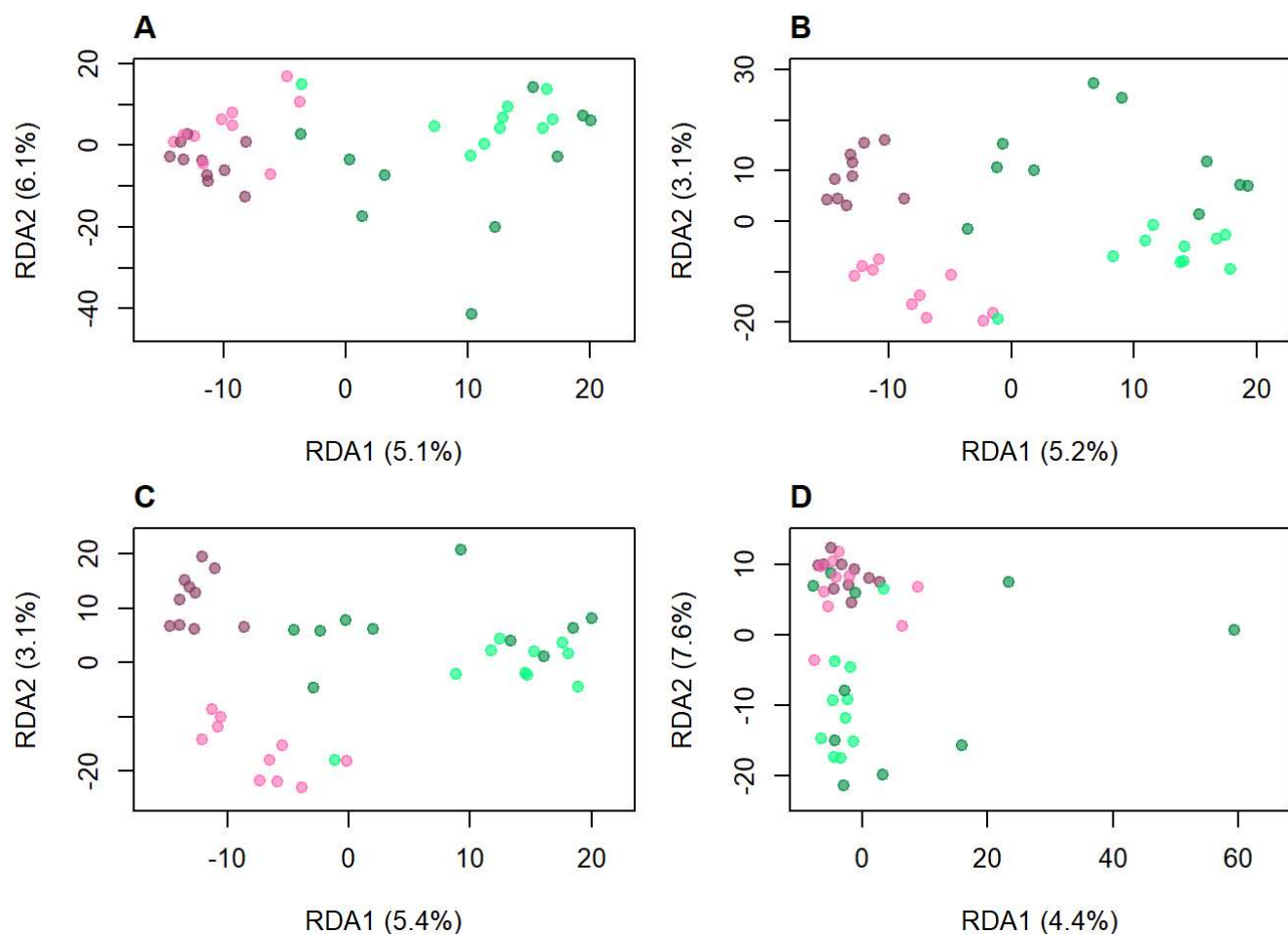


Fig. 2. Epigenetic differentiation plots. Ordination plots of the distance-based redundancy analysis (dbRDA) carried out to test for epigenetic differences in CG context with all four models: 1. $ED \sim M$ (A), 2. $ED \sim M * S$ (B), 3. $ED \sim M * S + \text{Condition}(PC)$ (C), 4. $ED \sim PC$ (D). RDA1: first constrained ordination axis; RDA2: second constrained ordination axis. Samples are colored by group (i.e.: maternal environment x real-time herbivory simulation).

CHG

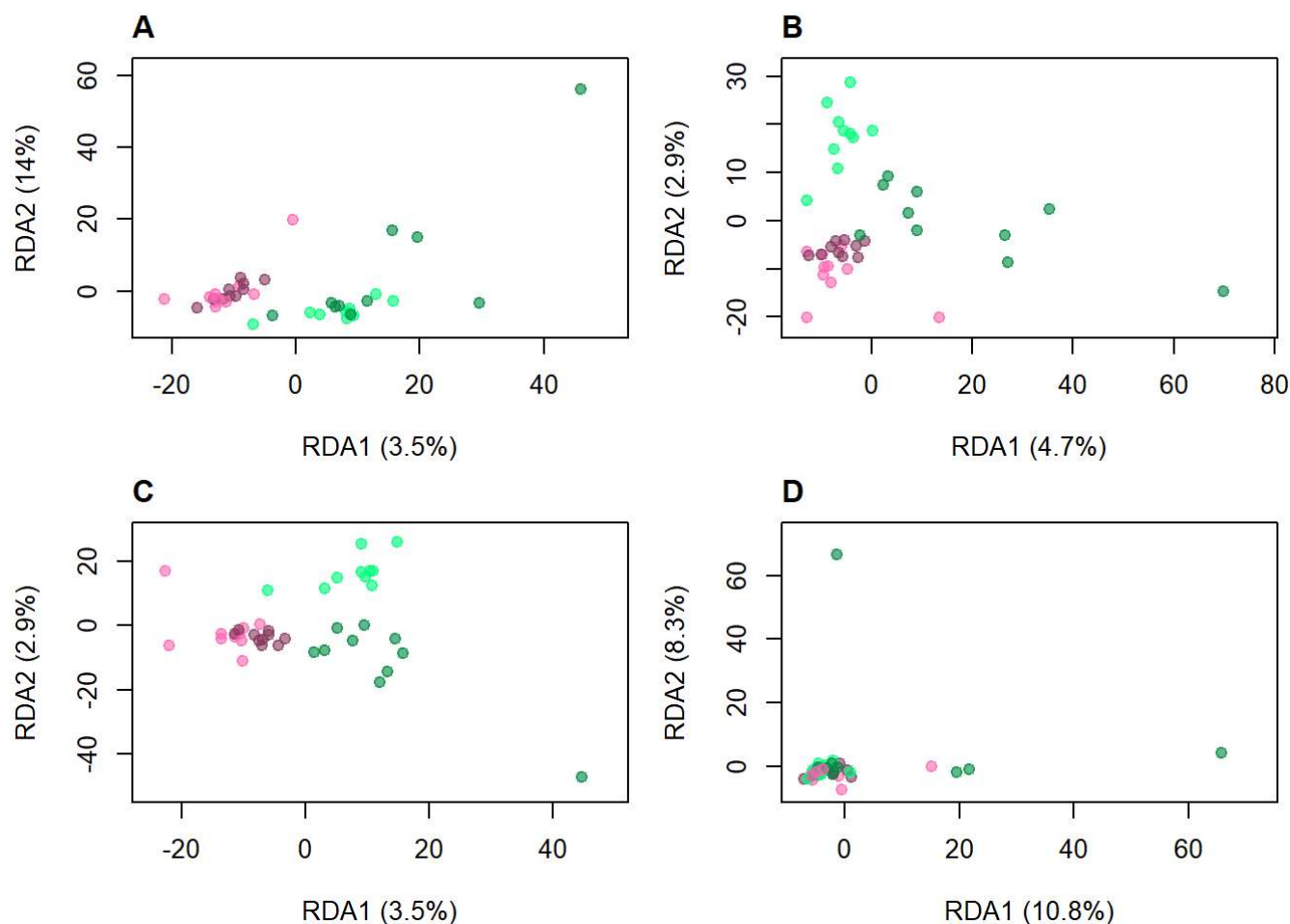


Fig. 2. Epigenetic differentiation plots. Ordination plots of the distance-based redundancy analysis (dbRDA) carried out to test for epigenetic differences in CHG context with all four models: 1. $ED \sim M$ (A), 2. $ED \sim M * S$ (B), 3. $ED \sim M * S + \text{Condition}(\text{PC})$ (C), 4. $ED \sim \text{PC}$ (D). RDA1: first constrained ordination axis; RDA2: second constrained ordination axis. Samples are colored by group (i.e.: maternal environment x real-time herbivory simulation).

CHH

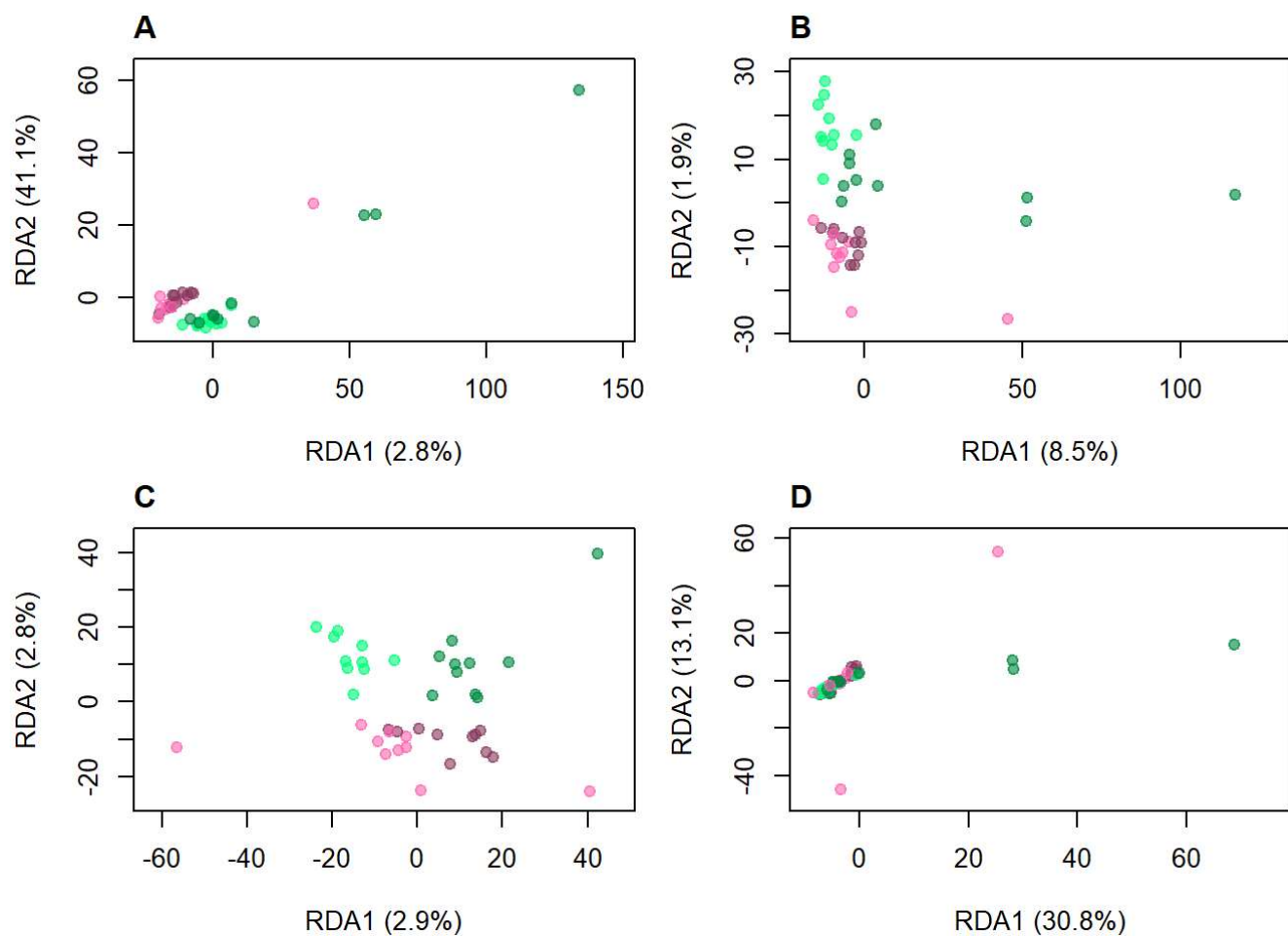


Fig. 3. Epigenetic differentiation plots. Ordination plots of the distance-based redundancy analysis (dbRDA) carried out to test for epigenetic differences in CHH context with all four models: 1. $ED \sim M$ (A), 2. $ED \sim M * S$ (B), 3. $ED \sim M * S + \text{Condition}(\text{PC})$ (C), 4. $ED \sim \text{PC}$ (D). RDA1: first constrained ordination axis; RDA2: second constrained ordination axis. Samples are colored by group (i.e.: maternal environment x real-time herbivory simulation).

all_CTXT

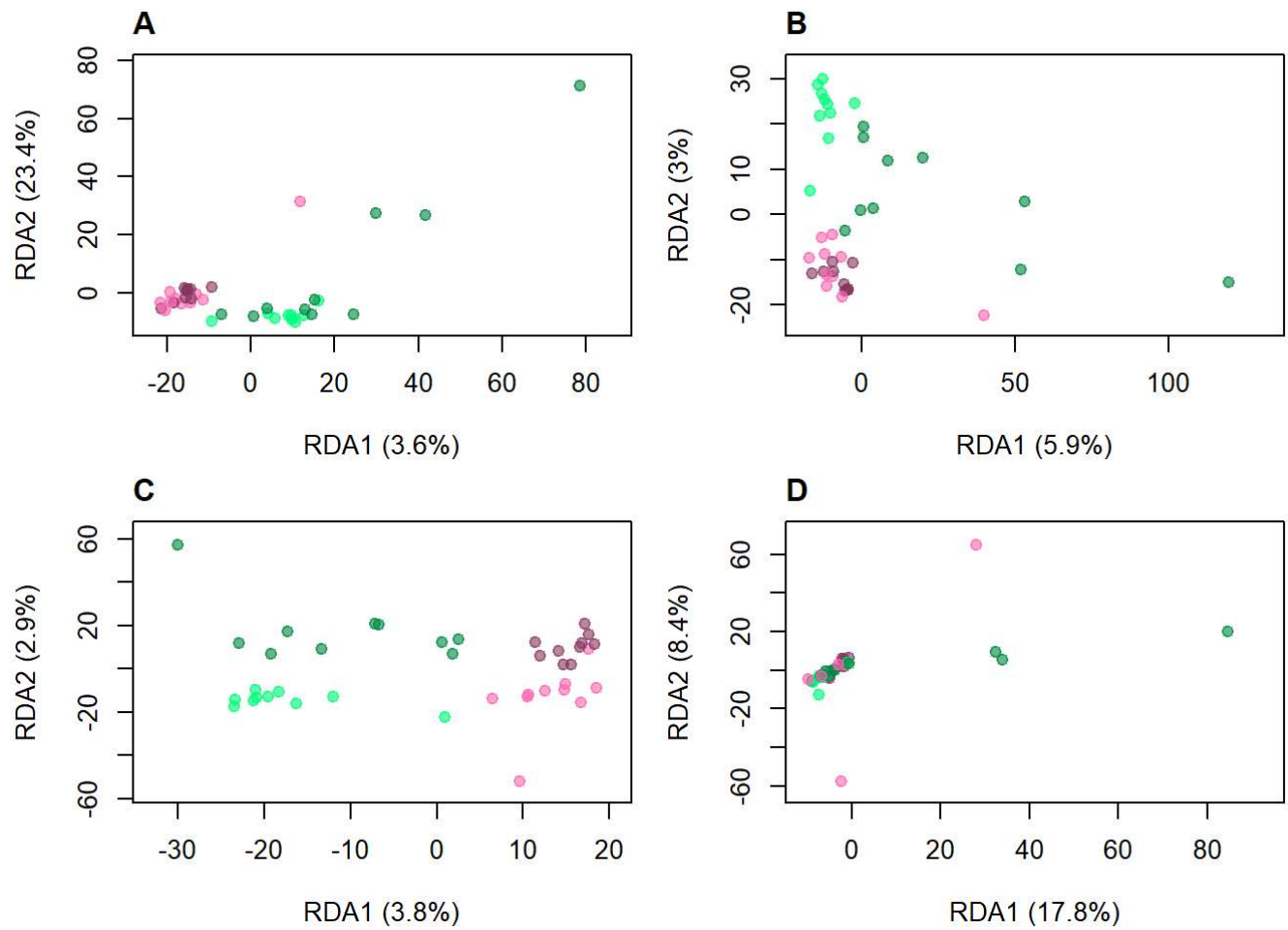


Fig. 4. Epigenetic differentiation plots. Ordination plots of the distance-based redundancy analysis (dbRDA) carried out to test for epigenetic differences in all_CTXT context with all four models: 1. ED ~ M (A), 2. ED ~ M * S (B), 3. ED ~ M * S + Condition(PC) (C), 4. ED ~ PC (D). RDA1: first constrained ordination axis; RDA2: second constrained ordination axis. Samples are colored by group (i.e.: maternal environment x real-time herbivory simulation).